Mapping Synesthetic Sequences in Space: Improving Consistency Test by Harnessing Cartography Tools

Optimizing Consistency Methods

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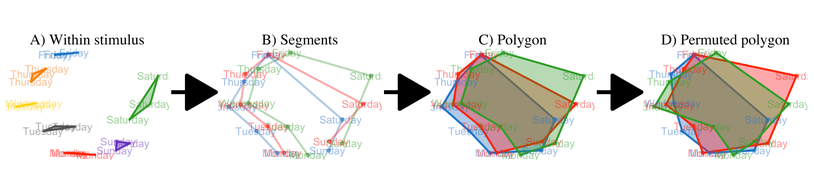
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Introduction

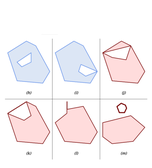
Sequence-Space Synaesthesia (SSS), is the phenomenon where people visualize ordered sequences in consistent spatial positions **(Flournoy, 1893; Galton, 1880)**. **Consistency tasks** classify SSS from controls by having participants indicate perceived spatial positions for a stimuli on a screen. Because of SSS internal spatial representations responses are more consistent than those of controls. Existing **consistency methods (i.e. Rothen et al., 2016)** test the spatial variability to the same stimuli, here we explored novel consistency methods.

Methods

Consistency tasks measure the responses (as a *x,y* screen position) for weekdays (Mon-Sun), months (Jan-Dec) and numbers (1-9), each repeated three times. The novel **consistency methods** *B)* and *D)* evaluate participant's *space-form* consistency (Figure 1).



Design	Method	Formula	Minimum Space-form	Maximum Space-form
Permuted valid structure [boolean/9]	D	$isValid_{perm}$		
Polsby–Popper [score/9]	C	$\frac{4\pi A}{P^2}$		
Perimeter [z-score]	A	P_{zs}		
Valid structure [boolean/9]	C	$isValid_{poly}$		
Permuted P-P [score/9]	D	$\frac{4\pi A_{perm}}{P^2_{perm}}$		
Segment self-intersections [n/9]	B	$SelfIntersect_{segm}$		
Area of the polygon [z-score]	C	A_{poly}		
Simple topology [boolean/9]	C	$isSimple_{poly}$		
Area [z-score]	A	A_{zs}		
Perimeter of the polygon [z-score]	C	P_{poly}		
Perimeter [screen size %]	A	$100\frac{A}{A_{screen}}$		
Area [screen size %]	A	$100\frac{P}{A_{screen}}$		



isValid evaluates if a polygon is topologically valid (i.e. blue polygons) or not (i.e. red polygons). Topologically valid is defined by the Open Geospatial Consortium **(Herring, 2010)** with the ***PostGIS 3.6.2dev Manual*** **(n.d.)** cartography tools.

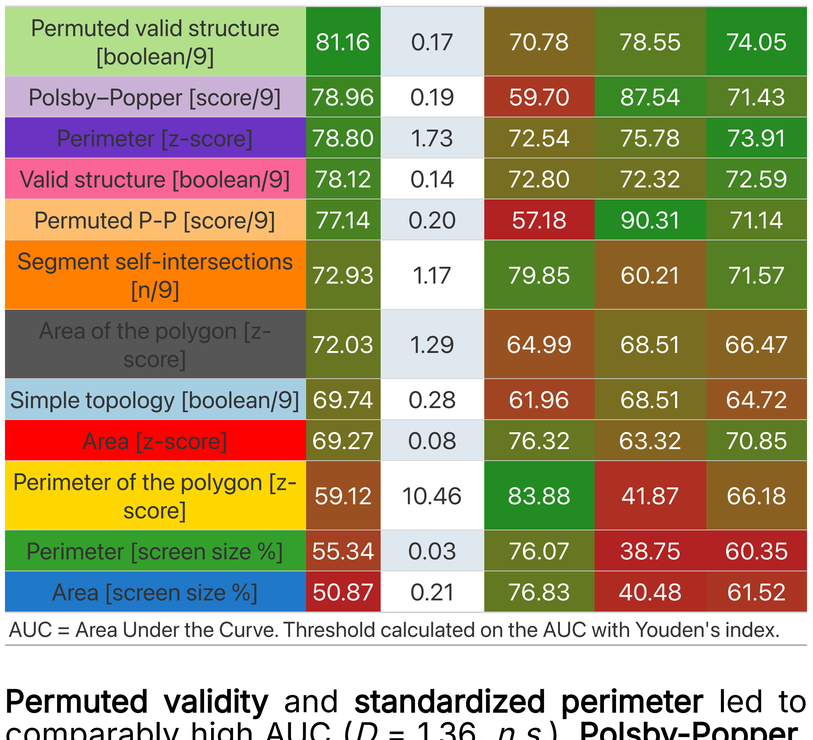
Aggregated datasets of consistency tasks

We aggregated four available datasets using the same consistency task **(Rothen et al., 2016; Van Petersen et al., 2020; Ward, 2022)** and Ward (2025, personal communication). Original groups are defined by self-report of having SSS or not (i.e. controls).

	SSS		Controls	
	N	Age	N	Age
Rothen (2016)	32	23.1	37	28.2
Ward (2022)	252	37.2	213	19.9
van Petersen (2020)	22	23.2	21	21.6
Ward (2026)	90	NA	18	NA
Total	396	NA	289	NA

Results

Receiver Operator Curves (ROC) of each consistency methods show the sensitivity (SSS classified as SSS) and sensibility (control as controls) by moving threshold for each consistency method.

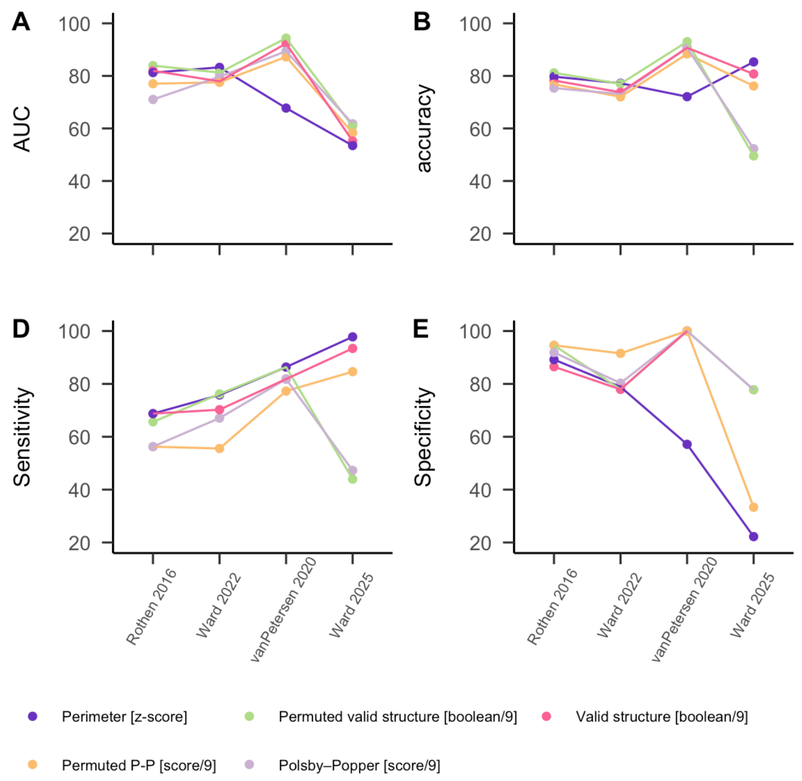


Feature	AUC	threshold	sensitivity	specificity	accuracy
Permuted valid structure [boolean/9]	81.16	0.17	70.78	78.55	74.05
Polsby–Popper [score/9]	78.96	0.19	59.70	87.54	71.43
Perimeter [z-score]	78.80	1.73	72.54	75.78	73.91
Valid structure [boolean/9]	78.12	0.14	72.80	72.32	72.59
Permuted P-P [score/9]	77.14	0.20	57.18	90.31	71.14
Segment self-intersections [n/9]	72.93	1.17	79.85	60.21	71.57
Area of the polygon [z-score]	72.03	1.29	64.99	68.51	66.47
Simple topology [boolean/9]	69.74	0.28	61.96	68.51	64.72
Area [z-score]	69.27	0.08	76.32	63.32	70.85
Perimeter of the polygon [z-score]	59.12	10.46	83.88	41.87	66.18
Perimeter [screen size %]	55.34	0.03	76.07	38.75	60.35
Area [screen size %]	50.87	0.21	76.83	40.48	61.52

AUC = Area Under the Curve. Threshold calculated on the AUC with Youden's index.

Permuted validity and **standardized perimeter** led to comparably high AUC (D = 1.36, *n.s.*). **Polsby-Popper**, differed from permuted validity (D = 2.06, $p < .05$).

However some results varied depending on the datasets:



Discussion

- Permuted validity underlines the possible role of topological rules for SSS space-forms, with the 0.17 cut-off reflecting an average of 1.53 valid forms out of 9.
- Polsby-popper score **(Polsby & Popper, 1991)**, originally designed to measure district compactness, performs surprisingly well in distinguishing SSS from controls, tough it's low sensitivity.
- Perimeter captures spatial variability within stimuli.

Earlier methods **(Rothen et al., 2016)** allowed controls to achieve artificially high consistency by clicking on the same position. While this limitation is adressed by novel methods. However, these interoduces a new issue: reduced detection of zig-zag SSS space-forms. Moreover, both methods lead to comparable classification performances. Overall, we find some heterogeneities in classification performances across datasets, motivating additional further valiadation in *Phase II* on yet-to-be collected data.

In sum, validity is as effective as previous methods for classifying SSS. Nevertheless, these results suggests the space-forms of SSS might be mapped following topographical rules such as for cartography.

References

This is a reproducible poster using the 'posterdown' package version 1.0 **(Thorne, 2019)**. Follow the QR code link for the code and data.

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